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ORIGINAL PAPER

Application of Frequency Ratio and Logistic Regression Model in the Assessment of Landslide Susceptibility Mapping for Nilgiris District, Tamilnadu, India

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Abstract This study aims to develop landslide susceptibility mapping for Nilgiris district of Tamil Nadu, India using GIS and remote sensing. The Northeast monsoons often trigger numerous landslides in the Nilgiris district of Tami Nadu, resulting in a high death toll and considerable property damage. The landslide causative factors namely elevation, slope, aspect, land use/land cover, rainfall, geology, geomorphology, soil, distance to lineament, distance to road, distance to stream, stream power index and topographic wetness index extracted from the spatial database were incorporated in the analysis, where the correlation between each parameter and landslides were computed using the frequency ratio model and logistic regression model. Finally, the resultant landslide susceptibility zonation maps were obtained from both the models, which were classified as very high, high, moderate, low and very low landslide susceptible zones. The results convey that about 8.78% and 23.22% of the study area were found to be prone to very high landslide susceptibility zones based on the frequency ratio and logistic regression model, respectively. The model validation was carried out using the Receiver Operating Characteristic (ROC) curve by correlating the information obtained from the field verification. The obtained Area Under the Curve (AUC) value of the frequency ratio and logistic regression model was 82.3% and 84.2%, respectively. Therefore, the landslide susceptibility index map of Nilgiris district is claimed to be valuable for the decision-makers to reduce the susceptibility associated with landslide hazard.

Keywords Landslide susceptibility mapping · GIS · Frequency ratio · Logistic regression · Nilgiris district

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Introduction

Landslides were a recurring phenomenon in India, which have to be addressed effectively to mitigate its adverse effects. The wide variation in the trend of the rainfall over the years in the country explicitly portrays the impact of adverse climate change. Nilgiris district of Tamil Nadu, part of the Western Ghats, is one of the four most landslide-prone zones in the country. Nilgiris district is well-known for its tourism. Its scenic beauty, attractive climate and ambience draw the tourists throughout the year. The effect of tourism is the principal reason for the rapid urbanization and infrastructure growth in the district.

There have been specific predictions about the climate change reported in the Fourth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC) [1], that, the predominant climate change would persist for many more centuries because of the greenhouse gases which have been emitted since into the atmosphere. The report also claims that the adverse implications of the climate change would not be stopped even if any mitigation efforts are proposed for the reduction in the dissemination of greenhouse gases. These claims make a realization of the fact that the adverse climate effect may lead to localized torrential rainfall which may lead to a higher rate of landslide events [2].

There exist various human interventions in the region that serves to be the primary source of landslide; it includes clearing of vegetation, construction and infrastructure development on the slopes, which removes the natural way for the rainwater to drain. As a result, the rainwater permeates into the pores of the soil, loosens it, and also starts seeping out through a small crevice under massive rocks. The infiltration of rainwater into the soil increases the pore water pressure, due to which the soil loses its stability and reaches a critical stage [3]. When this happens, often the entire rock is brought down by the pore water pressure causing an avalanche or debris flow. Therefore it could be concluded that the cutting of slope at the toe, substantial loading on the slope at the crest, hindrance in the surface drainage system and choking of weep holes in retaining structures, improper planning and design/implementation of remedial measures, vegetation removal and dumping are possibly the primary factors for most of the landslides in Nilgiris during intense rainfall conditions.

Concerned about the prediction of landslide occurrence, many studies have been carried out under various domains. Clubbing it into two broad classes, the first is the identification of prediction changes in the rainfall pattern and performing analysis based on the rainfall-induced behavior of landslides; the other category encompasses the assessment of landslide susceptibility using the GIS technique incorporating the statistical models.

Several works have been carried out to delineate the hazard-prone zones all over the earth. As many techniques exist, researchers have adopted various methodologies and techniques to bring out promising solutions. The methods are broadly classified into quantitative and qualitative techniques. The frequency ratio method takes a statistical approach quantitatively. Any system used to generate the hazard zonation map involves the consideration of the causative factors inducing landslides. The association between landslide areas and its accompanying factors can be brought out by analyzing the relations between the region with/without past landslides and the landslide-inducing factors [4]. However, the logistic regression model is also a statistical approach involving the multivariate analysis useful in the prediction of the presence or absence of characteristic or outcome-based values of a set of predictor variables [5].

The current study involves the frequency ratio model and logistic regression model for the assessment of susceptible landslide zone in the Nilgiris region and thereby validating and comparing the results of both models based on the field data obtained from previously occurred landslide zones and scarps. The probabilistic frequency ratio method had been widely adopted by various researchers in particularly for delineation of the landslides susceptible zones. Different landslide-prone areas have been studied using the frequency ratio method as it is a simple and comprehensible probabilistic model grounded on the observed relationship between the distribution of landslides and each landslide conditioning factor [6]. Over a decade some works have been published on frequency ratio model [2, 6–21]. On the other hand, logistic regression is said to be a more reliable statistical approach [22]. It is one of the most effective mathematical methods which are useful to find out the relationship between landslide causative factors and landslide locations [23].

Few researchers have worked on the Nilgiris study area and have produced promising results. However, this work differ from them based on the number of the causative factors included in the study. Guru Balamurugan et al. (2017) has opted ten factors for computing the landslide susceptibility zonation (LSZ) map using analytic hierarchy process and frequency ratio techniques, and the entire district hasn't been included in the study. Rajakumar et al. [24] have included only six factors to compute the susceptibility map of the Ooty town of Nilgiris district using overlay analysis. Saranaathan et al. [25] have opted only FR technique to produce the LSZ map opting seventeen parameters, however, their study area included only a smaller road (SH 37 Ghat road) of the Nilgiris district. Compared to the other works, we have opted thirteen prominent parameters and also compared Frequency ratio (FR) and Logistic regression (LR) technique which haven't been opted for this study area, moreover, the obtained results also prove promising to aid in disaster mitigation measures. Likewise, the parameters taken into account for this particular region varies when compared with other studies. Concerning with this particular study area, the parameters which are considered to be crucial in the occurrence of landslides are elevation, slope, aspect, land use/land cover, rainfall, geomorphology, geology, soil, distance to lineament, distance to road, distance to stream, stream power index and topography wetness index.

The primary drive of this research to execute research over the landslide-prone area of Nilgiris using the statistical frequency ratio model and logistic regression model for validating its performance. The prediction capability of the adopted model brings out an efficient hazard map which would serve as the best tool for hazard experts and urban planners to carry out their duty, thereby executing mitigation measures in Nilgiris area.

Study Area Description

The Nilgiris district of Tamil Nadu lies in 11.4916° N latitude and 76.7337° E longitude. The Nilgiris hill is a part of the Western Ghats; Doddabetta being the premier peak in South India with an altitude of 2,595 m, lies in this district. The geographical area of the district is 2,545 sq. km and holds a population of 7,35,394 as per the census of 2011. The district holds a natural forest which covers an area of 1,42,577 hectares that adds to the natural magnificence and economic wealth of the district. The gross area under cultivation is 75,505 hectares, and the major cropping involves mostly potato, coffee and various vegetable crops including cauliflower and cabbage along with spices such as cardamom, pepper and ginger, while tea being the major plantation crop as per the final estimates of 2015–16. The mean minimum temperature was 5 °C, and the mean maximum was 26 °C during 2015–16. The total rainfall during 2015–16 was 1357.7 mm. The prevalent weather and scenic beauty attract tourists throughout the year. Figure 1 shows the location of the study region.

Landslide Distribution in the Study Area

Landslides are a recurring phenomenon in India, that has to be addressed adequately to mitigate its adverse effects. The wide variation in the trend of the rainfall over the years in the country explicitly portrays the impact of adverse climate change. The urbanization activity includes the clearance of the vegetation cover, sturdy construction on slopes and often removal of the natural way for rainwater to drain. As a result, the rainwater permeates into the pores of the soil, loosens it, and also starts seeps out through a small crevice under massive rocks. The infiltration of rainwater into the soil increases the pore water pressure, due to which the soil loses its stability and reaches a critical stage. When this happens, often the entire rock is brought down by the pore water pressure causing an avalanche or debris flow. Therefore it could be concluded that the cutting of slope at the toe, heavy loading on the slope at the crest, blocking of the surface drainage system and choking of weep holes in retaining structures, improper planning and design/implementation of remedial measures, vegetation removal and dumping are possibly the primary factors for most of the

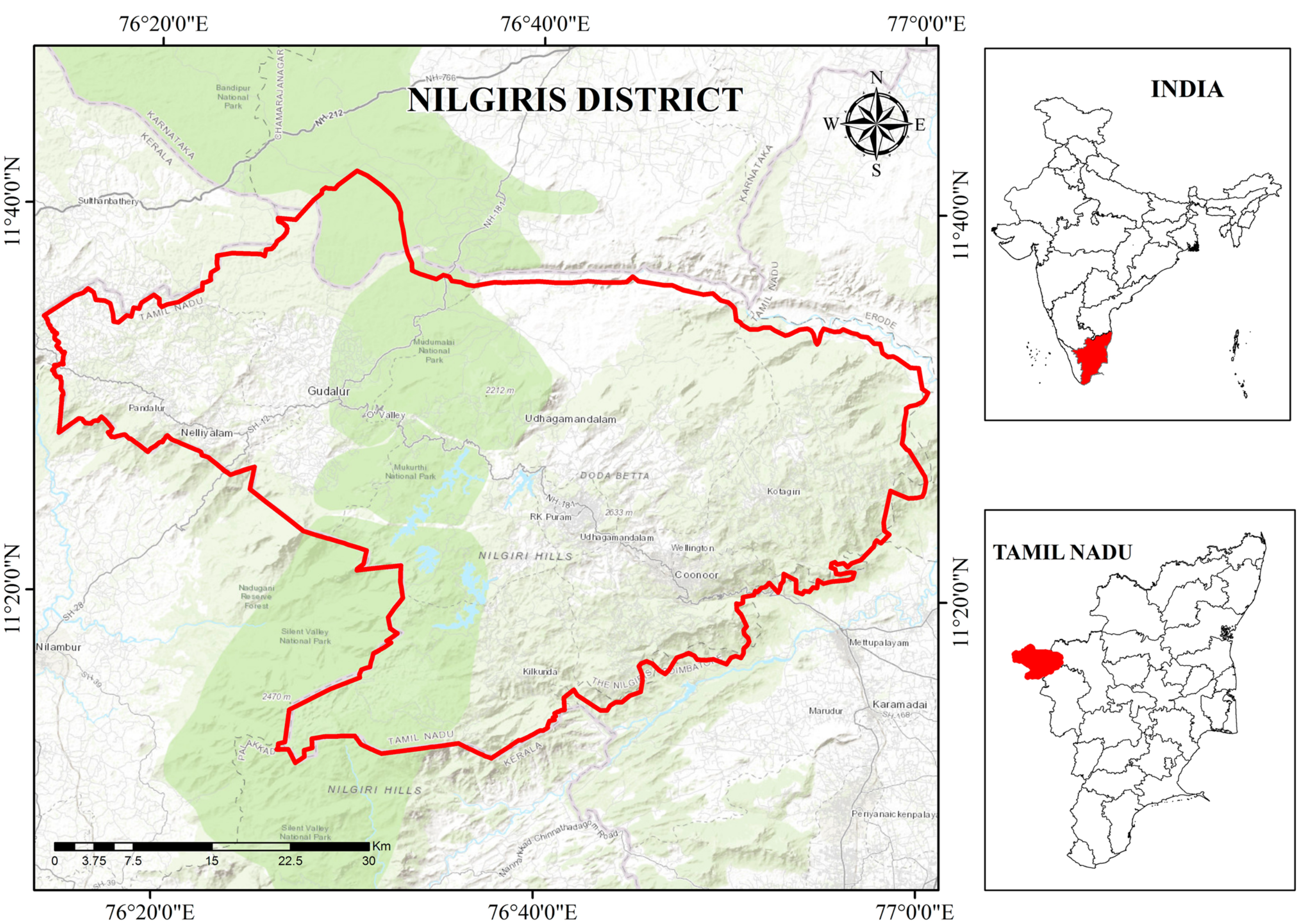

Fig. 1 Location map of the study area (Nilgiris District)

landslides in Nilgiris during intense rainfall conditions [26].

An analysis revealed that the 5-day antecedent rainfall (303 mm) and extreme rainfall on 10th November 2009 (405 mm) have saturated the slope thereby leading to a decrease in the matric suction and subsequent development of positive pore water pressure, which in turn has reduced the shear strength of the soil and resulted in shear displacement and a progressive failure. About 1150 landslides were reported within five days from 10 to 15 November 2009 causing 50 human lives and also vast damage on houses, roads and railway lines [27]. Most of the slides have been reported to have occurred over the past landslide scars. The history holds a record of the sequential occurrence of landslides in the Nilgiris district from 1865 to 2017. On average, about 94% of landslides have occurred along the cut slopes and 6% in the natural slopes. As per historical records, the majority of failures (97.2%) have been categorized as shallow translational debris slides, and only 2.4% as debris flows. The inventory records claim that there had been at least one landslide event per year, except in 1995. The average rate of occurrence was 20 landslides per year, while significant activities happened during 1992, 2001 and 2006 and 2009. Figure 2 depicts the distribution of the sites which has undergone landslides.

Database of Landslide Conditioning Factors

The landslides are triggered by natural phenomena like rainfall, earthquake or volcanoes. Though the triggering factors are naturally occurring ones, human interventions also trigger the occurrence of landslides. The physical parameters that highly influence the landslides are accounted for the present study. They include elevation, slope, aspect, land use land cover, rainfall, geology, geomorphology, soil, distance to lineament, distance to road, distance to stream, stream power index and topography wetness index. Using the frequency ratio model and logistic regression, the influence of each factor has been incorporated in the susceptibility mapping. Figure 3 portrays the causative parameter maps.

Elevation

It is a well-known fact that the landslides are more prominent in the elevated and undulated terrains; this proves evident that the elevation factor is likely to take part in the occurrence of landslides. However, the current research has incorporated the elevation parameter along with the slope gradient to delineate the flatlands from the undulated ones. Another reason which holds the importance accounting elevation is that the topographic attributes

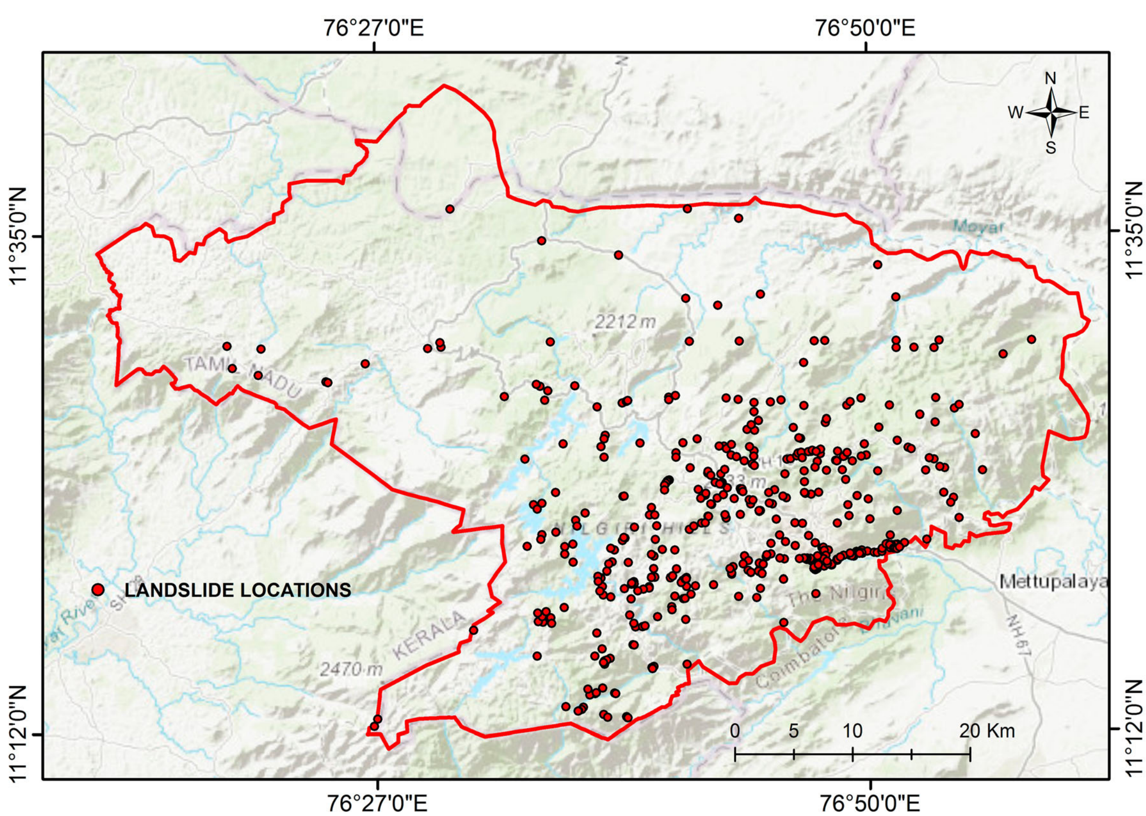

Fig. 2 Location of pre-occurred landslides in the study area

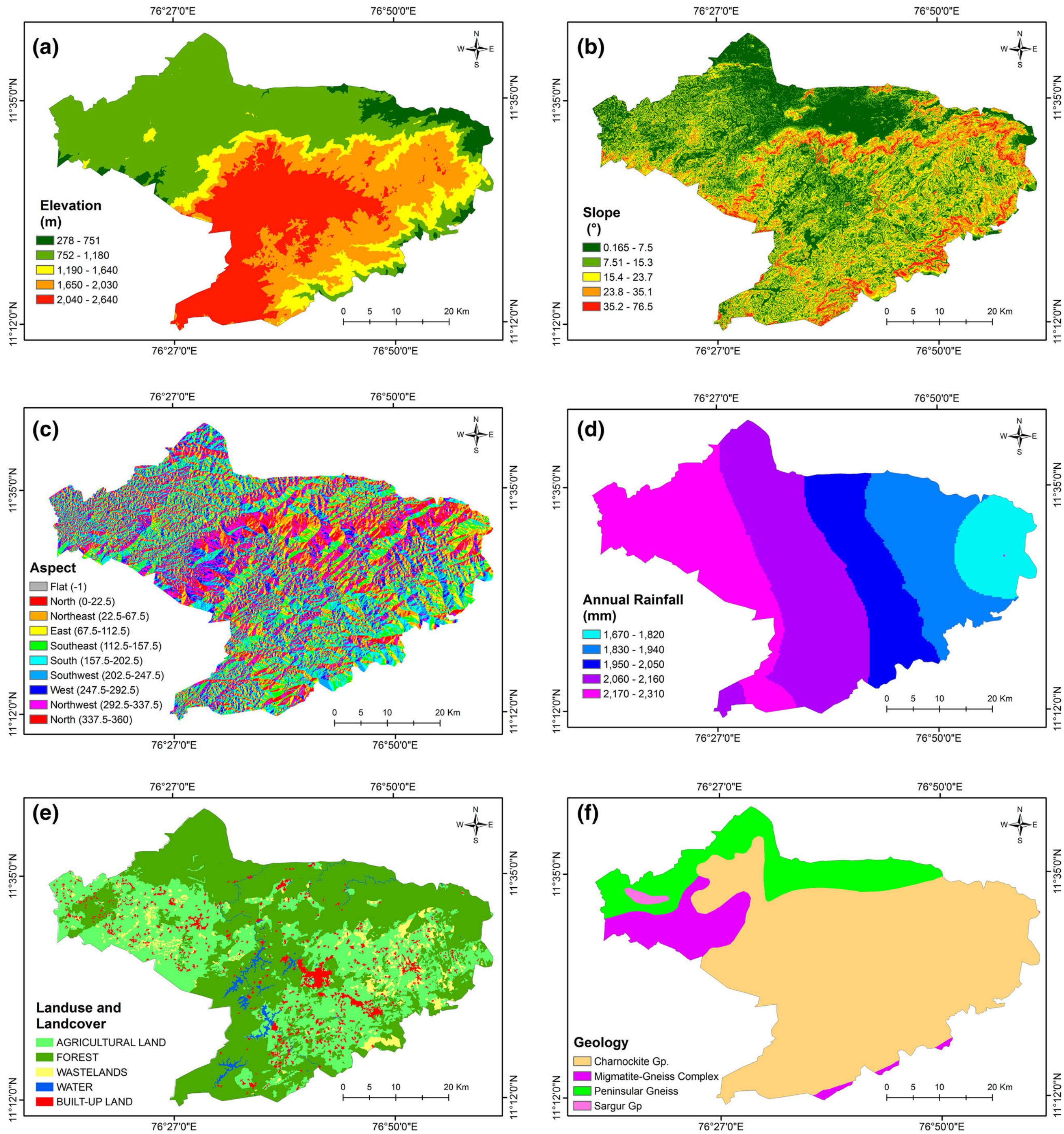

Fig. 3 Causative parameters of landslides **a** Elevation, **b** Slope, **c** Aspect, **d** Annual Rainfall, **e** Land use/land cover, **f** Geology, **g** Geomorphology, **h** Distance to lineament, **i** Distance to road, **j** Distance to stream, **k** Stream Power Index (SPI), **l** Topographic Wetness Index (TWI), **m** Soil

vary along with the landscape processes concerning the elevation; this can be found evident in the spatial distribution of the vegetation cover, and their types as the elevation vary. Figure 3a depicts the elevation map of the Nilgiris district, and its value ranges from 278 to 2640 m.

Slope

The rainfall-induced landslides are the prominent ones in the Nilgiris district. The slope is a primary factor that governs the runoff of the precipitation, thereby serving to be a critical parameter in the study of landslides. The slope angle and the friction angle between the materials are

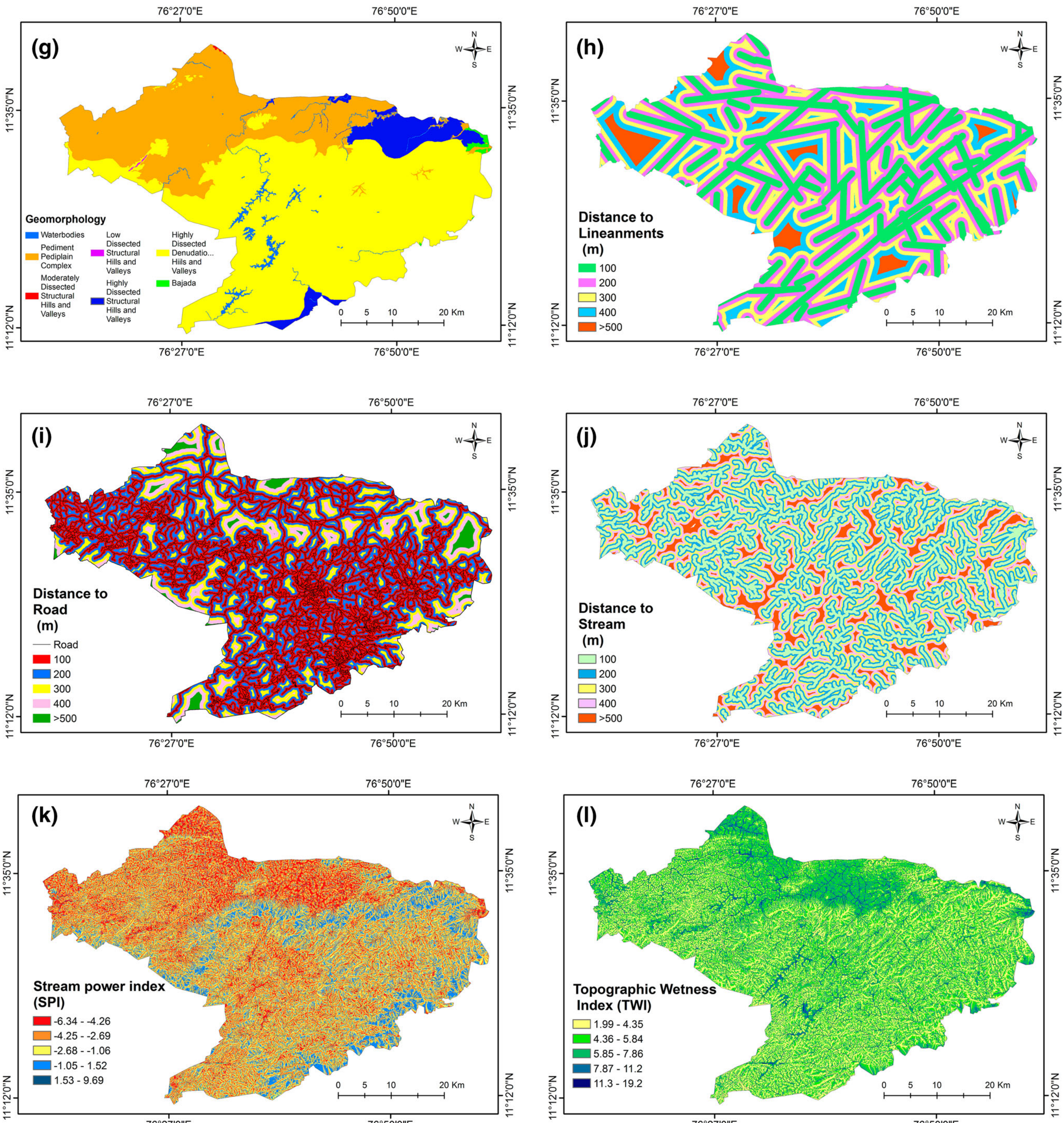

Fig. 3 continued

affected by the permeability and cohesion factors, which thereby controls the stability of the slope. Considering the slope angle in the study enables the identification of the unstable areas in the study area. All the reviewed literature have incorporated the slope angle in their study, and there also exists a claim that, the zone in mid-slope of mountainous terrain where relatively weak rocks like sandstone, mudstone and tuff outcrops as a single unit are highly prone to landslides [28]. Figure 3b depicts the slope map of the Nilgiris district, and its value ranges from 0.165 to 76.5°.

Aspect

The aspect which is defined here as the position of the terrain in a particular direction is taken into account.

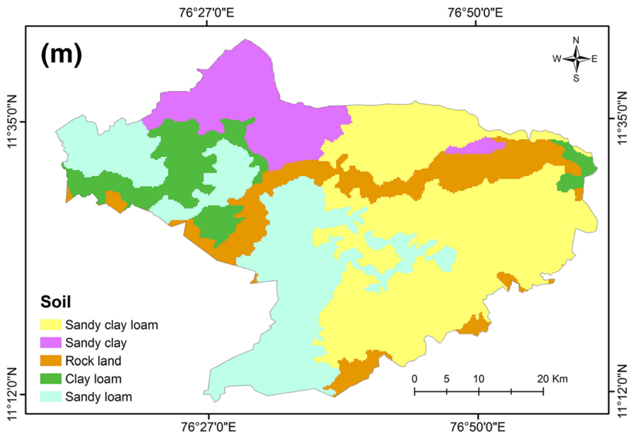

Fig. 3 continued

History of landslides ascertain that landslides have always occurred in a particular direction; thus the aspect is considered as a causative factor [29]. Though the aspect of the slope does not directly affect the slope stability, it affects the hydrologic processes such as the direction of precipitation and the amount of sunshine thereby determining the weathering process of the landforms. Usually, the aspect is accounted for all eight directions (North, Northeast, Northwest, South, Southeast, Southwest, East, West) along with flat terrain which has zero aspect ratio. Figure 3c depicts the aspect map of the Nilgiris district.

Rainfall

The Nilgiris district is prone to rainfall-induced landslides. Thus rainfall intensity claims to be a crucial factor for the mapping of susceptible zones, and it is necessary to consider the triggering factor in the study of landslides. Rainfall is the active parameter, and it influences the stability of the slope directly. The short span rainstorm or a series of continuous high-intensity rainfall throughout a prolonged period controls the runoff and energizes the pore water pressure, thereby causing loosening of landmass and affecting the stability of the terrain. Furthermore, the Nilgiris district is a critical region which receives heavy rainfall during both the Northeast and Southwest monsoons. Figure 3d depicts the average annual rainfall map of the Nilgiris district, and its value ranges from 1670 to 2310 mm.

Land use/land cover

The human interventions toward the natural habitat urge the susceptibility to disaster. These interventions could be well understood when studying the land use/land cover of the region. A hilly terrain has distinguished topographic character where the landforms vary concerning the elevation, slope and weather conditions. A complete understanding of the utilization of the landforms is necessary to consider for assessing the susceptible landslide zone. In the Nilgiris district, over-population and urbanization have led to deforestation, overgrazing, intensive agriculture and urbanization. Moreover, the developments and utilities have been carried out by cutting the slopes, thereby depreciating its stability. The forest cover is a significant land cover which plays a vital role in controlling soil erosion but, the interaction between the forest cover and the landslide activity is very complicated and crucial to understanding. Figure 3e depicts the land use/land cover map of the Nilgiris district and its classified into agricultural land, forest, wastelands, water and built-up land.

Geology

The Nilgiris district belongs to the Archean continental landmass of the Indian peninsula which is composed of mainly metamorphic rocks (gneisses, charnockites and crystalline schist). The underlying rock or the bedrock serve to be influential in the stability of the slope. The reaction of the water with the rocks and the behavior of the rock mass concerning the pore water pressure have to be understood to categorize the critical rock type, which triggers the landslide activity. Figure 3f depicts the geology map of the Nilgiris district and its classified into charnokite, migmatite-gneiss complex, peninsular gneiss and sargur.

Geomorphology

The earth's surface is profoundly influenced by the surface processes viz. physical, chemical and biological process which operates on or near the earth's surface, which creates various topographies or landforms. The interaction of the climate, hydrologic and biologic phenomenon with the geology determines the geomorphology of the region. Therefore, the geomorphology serves to be an essential factor in the study of landslides as it portrays the implication of the most dominant features which influences the earthly phenomenon. Rigid type structural hills are predominant in the Nilgiris district. Figure 3g depicts the geomorphology map of the Nilgiris district and its classified int waterbodies, pediment-pediplain complex, low, moderately and highly dissected structural hills and valleys, highly dissected denudational hills and valleys, bajada.

Distance to lineament

The landslides are highly correlated with the tectonic fractures and the scarps that are prevalent in the earth surface. The lineaments expose the underlying geological structure like faults as a linear feature in a landscape. The shear and fracture zones, as well as igneous intrusions like dykes, are also expressed as geomorphic lineaments. The lineaments can be viewed obviously in the satellite imageries or airborne photographs. The lineaments are liable to the occurrence of landslides, and as the distance from the lineament increases the liability decreases; thus included as a causative parameter in the study. The faults are the ones that make the rocks highly fractured, thereby reducing its strength and stability. Figure 3h depicts the distance to lineament map of the Nilgiris district and its classified into five classes with 100, 200, 300, 400 and > 500 m distance from the lineament lines.

Distance to road

The urbanization involves the construction of the utilities and enhancement of infrastructure. In the hilly terrain like Nilgiris, the construction of the road or any utility on the slopes cause a hike in the stress on the back of the slope thereby creating tension cracks. These vitally reduce the stability of the slope causing damage to the toe support. As the construction of roads influences the stability of the slope, the distance from the road is accounted for the varying loss in the stability due to the construction activity. Figure 3i depicts the distance to road map of the Nilgiris district and its classified into five classes with 100, 200, 300, 400 and > 500 m distance from the roads.

Distance to Stream

The streams and the rivers play a vital role in influencing the saturation degree of the soil, which has a direct link to the pore water pressure, which is a crucial factor that affects the slope stability. The streams are active agents that erode the materials, which also impacts the slope stability. Moreover, construction of any drainage structure by the streamside intensively affects the stability. These above-mentioned reasons paved the way to consider the distance from the stream streams and its susceptibility to landslides. Figure 3j depicts the distance to stream map of the Nilgiris district and its classified into five classes with 100, 200, 300, 400 and > 500 m distance from the streams.

Stream Power Index (SPI)

The stream power index is calculated based on the local slope gradient and the catchment area. The potential flow erosion at a given point can be described using the SPI, and it is directly proportional to the erosion risk. The increase in the catchment area and the slope gradient indicates a prominent increase in the SPI, which denotes the increase in the amount of water contributed to the up-slope region with the increased flow velocity. The SPI is calculated as:

$$\mathrm{SPI} = \ln\,(\mathrm{CA}\ .\ \tan\,\mathrm{G}) \tag{1}$$

where CA denotes the catchment area, and G denotes the slope gradient. Figure 3k depicts the SPI map of the Nilgiris district, and it ranges from -6.34 to 9.69.

Topographic Wetness Index (TWI)

The topographic wetness index quantifies the topographic control on hydrologic processes. It is a steady-state wetness index and a secondary geomorphometric parameter that portrays the effect of topography on the location and size of the saturated source areas of runoff generation. Prominent disruption in the original slope surface during the movement of landmasses shows high TWI range values whereas for translational rockslides and slumps shows low TWI range values. The spatial pattern of the high TWI values depicts the prominent drainage lines on a landslide body. The TWI is determined as:

$$\mathrm{TWI} = \ln\,(a/\ \tan\beta) \tag{2}$$

where a implies the specific catchment area, and β depicts the slope gradient. Figure 3l depicts the TWI map of the Nilgiris district, and it ranges from 1.99 to 19.2.

Soil

The characteristics of the soil play a vital role in the stability of the slope. The soil parameters and their slope characteristics determine the occurrence and type of landslides. Soils having a shallow depth on steep slopes are susceptible to landslides [23]. Soil map was collected from the National Bureau of Soil Survey (NBSS). Figure 3m depicts the soil map of the Nilgiris district, and it is classified into five classes such as: sandy clay loam, sandy clay, rock land, clay loam and sandy loam.

Frequency Ratio Model

Frequency ratio model is a comprehensible and straightforward method to envisage the landslide susceptible zones. It adopts the bivariate statistical method, and an underlying assumption is claimed that the occurrence of the landslide is dependent on the influencing factors. The central computation involved in the model is the correlation between the causative factors and the location of the landslide zones in the study area. This model differentiates the zones which hold a record of previous landslides and their conditioning factors from the zones which have never faced landslides and the prevalent landslide-related factors

Table 1 The spatial relationship between each landslide conditioning factor and landslide by Frequency Ratio model

Parameter	Class	Number of pixels in class	Percentage of pixels in the class (%; PIF)	Number of landslide pixels	Percentage of landslide pixels (%; PLO)	Frequency Ratio (FR)
Aspect	Flat	18,609	0.680622183	151	0.395868289	1.719314738
	North	201,336	7.363842646	2395	6.278838087	1.172803398
	Northeast	344,802	12.61109624	4639	12.16180789	1.036942563
	East	332,504	12.1612982	4816	12.62583893	0.96320714
	Southeast	360,894	13.19965941	5697	14.93550755	0.883777091
	South	348,753	12.75560364	5594	14.66547819	0.869770728
	Southwest	302,114	11.04978721	4486	11.76069631	0.939552125
	West	278,823	10.19792138	3749	9.828544463	1.037582056
	Northwest	351,459	12.8545753	4306	11.28880034	1.138701626
	North	194,822	7.125593793	2311	6.058619966	1.176108393
	SUM	2,734,116	92.87440621	38,144	100	0.928744062
Slope	0.165–7.5	761,457	27.8559035	8160	21.39261745	1.30212694
	7.5–15.3	830,717	30.38959861	16,101	42.2110948	0.719943388
	15.3–23.7	640,822	23.44278901	9960	26.11157718	0.897792916
	23.7–35.1	375,435	13.73430296	2922	7.660444631	1.792885873
	35.1–76.5	125,126	4.577405922	1001	2.62426594	1.744261453
	SUM	2,733,557	100	38,144	100	1
Geology	Charnockite Gp	2,052,672	75.09161141	36,886	96.70197148	0.776526169
	Migmatite-Gneiss Complex	234,352	8.573152124	1002	2.626887584	3.263615914
	Peninsular Gneiss	431,125	15.77157528	256	0.67114094	23.49964716
	Sargur Gp	15,408	0.563661193	0	0	0
	SUM	2,733,557	100	38,144	100	1
Geomorpholgy	Waterbodies–unclassified	48,598	1.777830131	676	1.772231544	1.003159061
	Highly Dissected Structural Hills and Valleys	165,197	6.043298164	296	0.776006711	7.787688013
	Moderately Dissected Structural Hills and Valleys	896	0.032777806	0	0	0
	Bajada	9820	0.359238896	0	0	0
	Pediment-pediplain Complex	842,494	30.82042921	1013	2.655725671	11.60527593
	Highly Dissected Denudational Hiils and Valleys	1,665,501	60.92797772	0	0	0
	Low Dissected Structural Hills and Valleys	1051	0.038448073	0	0	0
	SUM	2,733,557	100	1985	5.203963926	19.21612091
Rainfall	1,670–1,820	248,336	9.084720019	843	2.210046141	4.110647217
	1,830–1,940	490,744	17.95257973	9227	24.18991191	0.742151513
	1,950–2,050	451,439	16.51470959	16,808	44.06459732	0.374784081
	2,060–2,160	869,608	31.81232365	9555	25.04981124	1.269962609
	2,170–2,310	673,430	24.63566701	1711	4.485633389	5.492126723
	SUM	2,733,557	100	38,144	100	1
SPI	− 6.34– − 4.26	639,208	23.38374506	5589	14.65236997	1.5959019
	− 4.26– − 2.69	969,947	35.48296231	13,850	36.30977349	0.977228963
	− 2.69– − 1.06	748,794	27.39266092	12,158	31.87395134	0.859405871
	− 1.06–1.52	301,605	11.03342641	4989	13.07938339	0.843573897
	1.52–9.69	74,003	2.707205301	1558	4.084521812	0.662796142
	SUM	2,733,557	100	38,144	100	1

Table 1 continued

Parameter	Class	Number of pixels in class	Percentage of pixels in the class (%; PIF)	Number of landslide pixels	Percentage of landslide pixels (%; PLO)	Frequency Ratio (FR)
TWI	1.99–4.35	1,542,601	56.43200416	22,710	59.53754195	0.947838999
	4.35–5.84	892,603	32.6535353	11,953	31.33651426	1.04202832
	5.84–7.86	210,010	7.682664016	2653	6.955222315	1.104589281
	7.86–11.2	74,108	2.71104645	667	1.748636745	1.550377148
	11.2–19.2	14,235	0.52075007	161	0.422084732	1.233757185
	SUM	2,733,557	100	38,144	100	1
LULC	AGRICULTURAL LAND	982,596	35.94569274	21,824	57.2147651	0.628259028
	FOREST	1,453,946	53.18879394	9480	24.85318792	2.140119574
	WASTELANDS	128,801	4.71184614	1557	4.081900168	1.154326649
	WATER	35,268	1.290187108	312	0.81795302	1.577336444
	BUILT-UP LAND	128,006	4.682763154	4976	13.04530201	0.358961651
	SUM	2,728,617	99.81928308	38,149	100.0131082	0.998062003
Distance to lineament	100	885,920	32.40905531	16,367	42.90845218	0.755307024
	200	813,837	29.77208816	12,717	33.3394505	0.892998766
	300	608,473	22.25938585	6606	17.31858221	1.285289152
	400	307,835	11.26133459	1903	4.988989094	2.257237765
	500	117,492	4.298136092	551	1.444526007	2.975464666
	SUM	2,733,557	100	38,144	100	1
Distance to stream	100	767,075	28.06142327	13,988	36.6715604	0.765209415
	200	681,703	24.93831297	9313	24.41537332	1.021418458
	300	603,388	22.07336448	6764	17.73280201	1.244775894
	400	452,958	16.57027821	5777	15.14523909	1.094091556
	500	228,201	8.348133951	2302	6.035025168	1.383280719
	SUM	2,733,325	99.99151289	38,144	100	0.999915129
Distance to road	100	1,411,862	51.64926138	30,963	81.17397232	0.636278599
	200	744,526	27.23652735	5359	14.04939178	1.938626795
	300	362,517	13.26173188	1129	2.959836409	4.48056245
	400	166,335	6.084928904	270	0.70784396	8.596426967
	500	48,317	1.767550485	423	1.108955537	1.593887605
	SUM	2,733,557	100	38,144	100	1
Elevation	278–751	145,133	5.309309446	1479	3.877411913	1.369292086
	751–1,180	1,077,584	39.42057912	3275	8.585885067	4.591323878
	1,180–1,640	314,606	11.50903383	4702	12.32697148	0.933646505
	1,640–2,030	616,092	22.53810694	17,533	45.96528943	0.490328838
	2,030–2,640	579,996	21.21762963	11,155	29.24444211	0.725526907
	SUM	2,733,411	99.99465897	38,144	100	0.99994659
Soil	Sandy clay loam	1,085,384	39.70592163	27,001	70.78701762	0.560920956
	Sandy clay	323,421	11.83150745	192	0.503355705	23.50526146
	Rock land	347,012	12.69452219	1364	3.575922819	3.549998933
	Clay loam	249,966	9.144349286	432	1.132550336	8.07412174
	Sandy loam	722,903	26.44550672	9136	23.95134228	1.104134641
	SUM	2,728,686	99.82180726	38,125	99.95018876	0.998715545

in it. The frequency ratio can be defined as the ratio of the probability of landslide occurrence to not occurrence for a given attribute [2]. The landslide susceptibility index (LSI) was calculated by a summation of each factor ratio value using the given formula equation

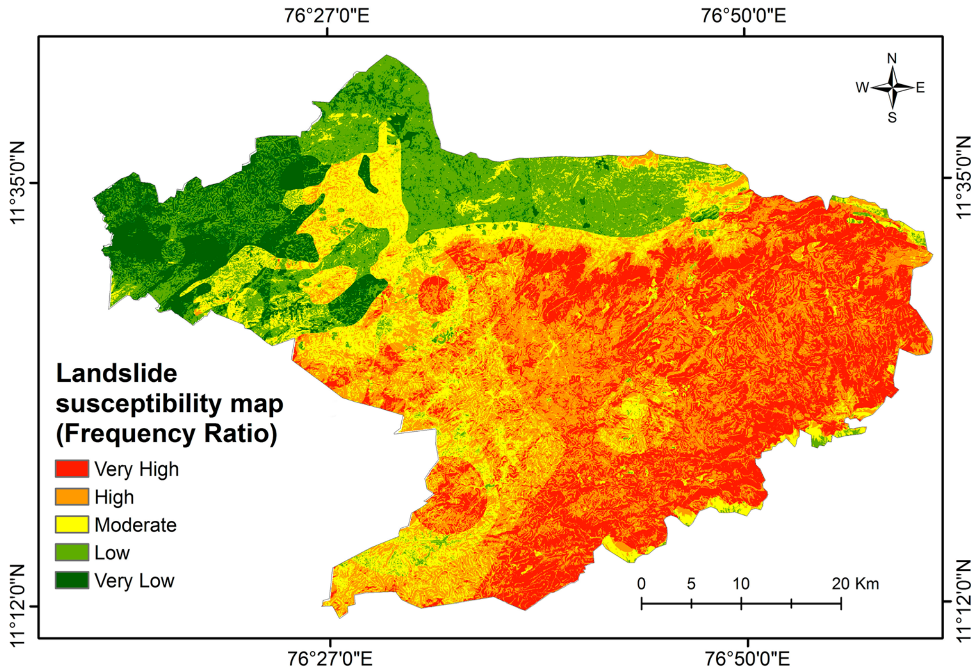

Fig. 4 Landslide susceptibility map of the Nilgiris district using Frequency ratio model

$$\mathrm{LSI} = \sum \mathrm{FR} \tag{3}$$

where LSI is the landslide susceptibility index, and FR is the frequency ratio of each factor type or class. Table 1 correlates the spatial relationship between landslide and each landslide conditioning factor by the proposed frequency ratio model.

The frequency ratio of value 1 is an average value for the areas where landslides have occurred in among the total area. If the FR value is higher than 1, it denotes the higher correlation which indicates a high probability of landslide occurrence and lower than 1 stated the lower correlation which implies a low probability of landslide occurrence. The thematic layers were generated using ArcGIS 10.3 software, and Microsoft Excel sheet was used to calculate the frequency ratio of each layer. The landslide susceptibility map was finally constructed using the ArcGIS software, and the resultant map was indexed with five susceptible zones (very high, high, moderate, low and very low) as depicted in Fig. 4.

Logistic Regression Model

Logistic regression model adopts the multivariate analysis approach, where one or more multivariate regression relation forms between a dependent and various independent variables. In the study, the dependent variable is a binary variable which represents the presence or absence of landslides and thus the logistic link function is applicable here. The logistic regression coefficients are used to estimate the ratio fo each independent variable in the model. Thus quantitatively, the relationship between the occurrence of landslides and its dependent factors can be expressed as:

$$p = \frac{1}{1 + e^{-z}} \tag{4}$$

where p is the estimated probability of landslide occurrence which varies from 0 and 1 on an S-shaped curve, and z is the linear combination which can be represented in the following form:

$$z = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n \tag{5}$$

where b_0 is the model intercept, b_i (i = 1,2,…,n) are the slope coefficients of the model, and x_i (i = 1,2,…,n) are the independent variables. This linear model is the logistic regression of the presence or absence of landslide occurrence on the independent variables [30]. A statistical package (R) has been used to assess the correlation between the landslide and each causative factor. The logistic regression formula (z_{13}) obtained from the model is as follows:

$$\begin{aligned} z_{13} = {} & Aspect * (0.0155076) + Stream * (-0.0011300) \\ & + Road * (-0.0039017) + Elevation * (0.0020344) \\ & + Geo\log y * (0.1115152) + Geomorpho\log y \\ & * (-0.2288318) \\ & + Lineament * (-0.0001406) + LULC * (-0.1431813) \\ & + Ra\inf all * (-0.0006693) + Slope * (0.0138038) \\ & + Soil * (-0.0828720) + SPI * (0.2281644) \\ & + TWI * (0.0938152) + (-1.0795450) \end{aligned} \tag{6}$$

where *Stream* is Distance to Stream; *Road* is Distance to Road; *Lineament* is Distance to Lineaments; *SPI* is Stream Power Index, and *TWI* is Topographic Wetness Index. The probability of landslide occurrence is thus computed finally from Eq. (4) to obtain the landslide susceptibility zonation map.

Results and Discussion

The proper handling of the causative factors using the frequency ratio model and logistic regression model delivers an appropriate landslide susceptibility zonation map. The emphasis is given to the chosen influential factors and their dependency on the occurrence of landslides. The results are thus entirely relied upon the behavioral impacts of the factors toward landslides. When concerned with the elevation, the SRTM 30 m Digital Elevation Model (DEM) was utilized; the Nilgiris district has a range of elevation between 278 and 2644 m. The DEM data were exploited further to compute the slope and aspect map.

The landslide-prone zone is found at the mid-altitude region of the hills; this invokes the inference that the highly sensitive region is not in the peaks of the hills instead they are in the mid part of the hilly terrain. The altitude along with the slope and aspect of the terrain is highly influenced by the weather, the direction of the rainfall and the direction of sun rays. Moreover, the increased urbanization can be found in the mid-elevated region compared to the fool hills and peaks. Likewise, the slope and the aspect of the terrain have been included in the analysis considering its impacts toward the hazard susceptibility.

It is evident from the Frequency ratio computed chart (Table 1) that the Northeast and southwest aspect is highly influential toward the occurrence of landslides. It is also evident that the landslides in the Nilgiris district are triggered by the rainfall which is predominant during the Northeast and Southwest monsoons; these seem correlated with the potential aspect ratio of the terrain. The slope attribute ranges up to 76° in the Nilgiris district. The slope influences runoff characteristics. Steeper the slope there exists no stagnation. At lower slope range, the infiltration rate is higher, because stagnation occurs there inducing an increase in pore water pressure. When a slope is subjected to increase in pore pressure due to infiltration or rise in the water table, the total and shear stresses remain essentially constant while mean effective stress decreases which correspond to a specific stress path which are examined under the critical state and instability [31]. The frequency ratio is thus higher in the table (1) for the lower slope ranges compared to the highly steeper ones.

In the case of the logistic regression model, based on the coefficients of each factor it is evident that SPI, Geology, TWI, Aspect and Slope considerably influence the probability of landslide occurrence due to its high weightage as given in Eq. (6). However, distance to stream, distance to road, distance to lineament and rainfall are found to have negligible influence in the occurrence of landslides.

The annual rainfall in the Nilgiris district ranges roughly from 1670 to 2310 mm. The district is a region which is prone to rainfall-induced landslides, and the region which

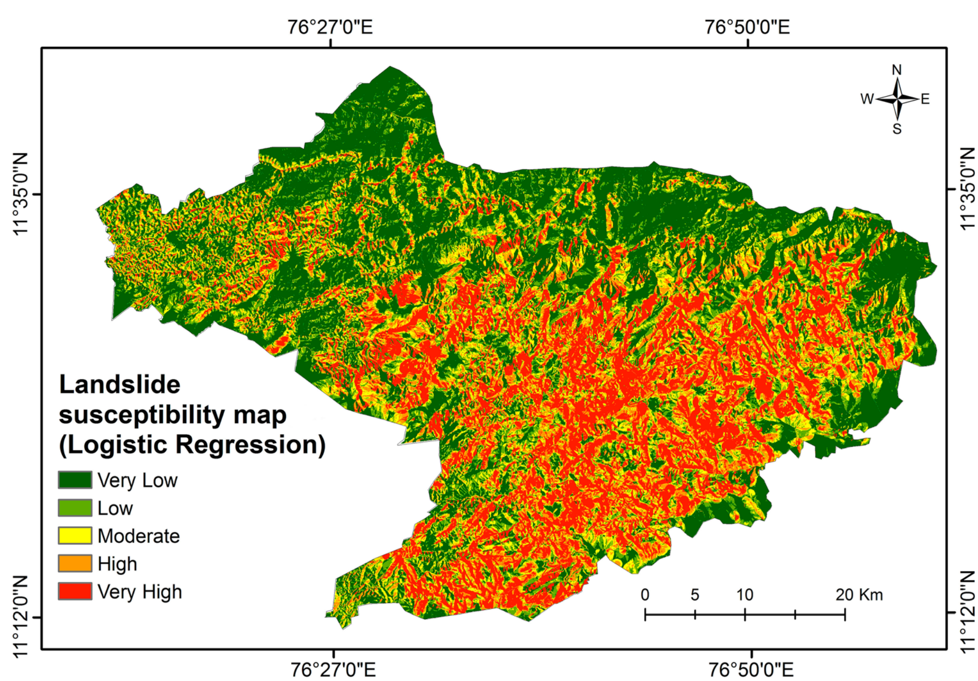

Fig. 5 Landslide susceptibility map of the Nilgiris district using Logistic Regression model

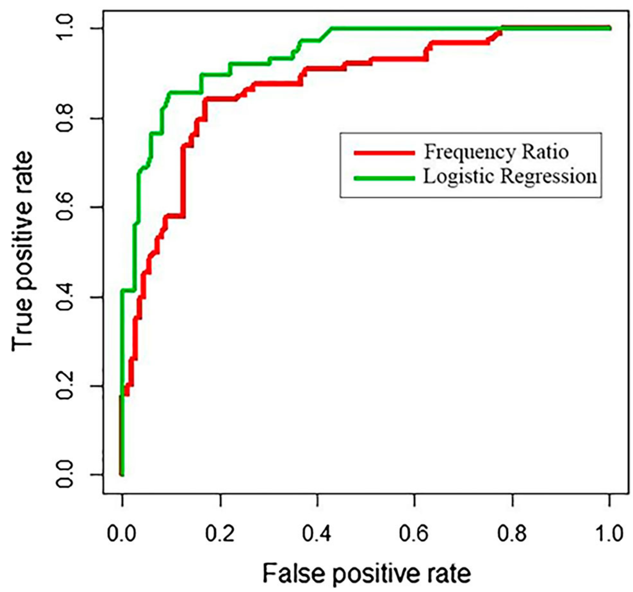

Fig. 6 ROC curve representing the quality of the model

receives steady rainfall of 2170–2310 mm is highly prone to landslides. The obtained results imply that there is no record of landslide in the region which receives the maximum rainfall, this may be due to the influence of the other causative factors. Though rainfall is heavy in a highly sloped region, the runoff would be predominant, saving it from a hazard zone. Likewise, the other contributing factors may play vitally to save a highly precipitated zone from being susceptible to landslide attack.

The water bodies and the highly saturated regions have a high increase in their pore water pressure, due to the substantial phenomenon of infiltration, making the zone susceptible to subsidence. The shrubs and plants in the land cover which do not have strong roots to hold the underlying landmass together make the landmass weaker. The dense vegetation cover with strong roots upholds the surface soil and prevent it from erosion. Moreover, the roots consume the surface water in dense forest thereby reducing the amount of water sucked up by the soil mass when compared to the open forests.

While taking geology and geomorphology into consideration, the syenite, the shales with bands of limestone and quartz vein are the types of minerals which are highly susceptible to landslides, as per the results obtained. This is because of their interaction with the infiltrated water; as they get saturated, they lose their stability and becomes vulnerable. The geomorphological landforms that seem susceptible are the hilltop weathered and the ridge type structural hills. Compare to valleys, the ridges are highly prone to landslides and, compared to the vast landmasses the weathered rocks are subjected to landslides.

The lineaments are the fractures that are prevalent on the surface due to the pressure underwent by the underlying landmasses. The region near the lineaments is viable to landslide occurrence. In the study, the buffer regions of 100, 200, 300, 400 and >= 500 were analyzed, by which it was understood that the region within 500 m from the lineaments are highly vulnerable. Likewise, the analysis was carried out over the buffered regions from roads. The vulnerability was found in the 300 and 400 m buffer, this evidence shows that the effect of construction reflects in a region which is nearly 0.4 km far. Similarly, the distance from the streams also was taken into account; which implied that, the closer the distance to streams higher the susceptibility to landslides.

The stream power index (SPI) between -6 to -4 is highly vulnerable to landslide attack; this implies that lower the SPI, higher the vulnerability; and as the SPI increases, the vulnerability decreases. While considering the topographic wetness index (TWI), it can be concluded that the higher the TWI, the higher the vulnerability. The wetness index thereby has a positive impact over the susceptibility of landslides which SPI influences negatively.

Each parameter is taken into consideration in this study influence the landslides variably; the frequency ratio model and logistic regression model serve to be a flexible one in understanding the variable differences and bringing out an optimal resultant hazard map. Figure 4, 5 depict the landslide susceptibility zonation map obtained from the frequency ratio and logistic regression model, respectively. From the resultant map obtained from frequency ratio model, we conclude that 8.78, 16.83, 18.27, 30.08 and 26.04% of the area in the Nilgiris district falls under the very high, high, moderate, low and very low susceptible landslide zones, respectively. Whereas, the logistic regression model concludes that 23.22%, 14.09%, 12.96%, 15.26%, 34.47% of the study areas fall under very high, high, moderate, low and very low landslide susceptible zones.

Validation of the Result

To visualize the performance of the model, the receiver operating characteristic curve has been adopted. This method estimates the overall accuracy of the opted model. The area under the ROC curve has been taken as a measure to denote the accuracy of the results obtained from the model. The ROC curve is depicted in 2-dimension with a true positive rate on the Y-axis and false positive rate on the X-axis. The top left corner is the ideal point of the false positive rate of zero and a true positive rate of one. However, the closer the value of the area under the curve (AUC) of the model comes to 1, the better is the quality of the model. There are five categories of AUC (area under the curve) value under the ROC curve such as excellent (0.90–1.00), good (0.80–0.90), fair (0.70–0.80), poor (0.60–0.70) and fail (0.50–0.60) to understand the accuracy level. Therefore, the models which hold higher AUC values are much preferred than the ones with lower values of AUC. The frequency ratio model and the logistic regression model, which were adopted to assess the landslide susceptibility of the Nilgiris district provided AUC value of 0.823 and 0.842, respectively, which falls under the 'good' category. It implies that both the model serve best in assessing the susceptible landslide regions of the study area. Figure 6 depicts the ROC curve of the frequency ratio and logistic regression model.

Conclusion

The disaster management studies gain importance in recent days as there exist a predominant increase in climate change over the years. The landslides have been focused in this study, to delineate the high landslide susceptible zones of the Nilgiris district which is one among the four landslide hazardous zones in India. The frequency ratio model and the logistic regression model are bivariate and multivariate statistical models, respectively, were opted in this study. Both these models are understandable and straightforward techniques for hazard mapping. The model incorporated thirteen causative parameters that take part in the occurrence of landslides and provided a qualitative result which when validated using the ROC curve. The resultant zonation maps provide the spatial distribution of landslides while it doesn't give any information on the time forecast of occurrence, degree of impact or the frequency of occurrence. Thus, the government authorities should take steps to mitigate the landslide effect. The generated resultant map would serve as the best indicator of landslide hazard zones for the Nilgiris district administration to keep notice that no development projects are supported in the susceptible zones in future. The urban development and disaster preventive measures therein can be carried out using this zonation map as primary data.

Declarations

Conflicts of interest

The atuthors have no competing interests.

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